Does a "Fuzzy Soil Index" Help a Robot Walk on Soft Ground?

A replication study of the *wcci2026_hunter* fuzzy soil model

Hunter humanoid / IsaacSim / HumanoidVerse

June 21, 2026

The problem (from the paper)

Goal: make a humanoid robot walk on **agricultural soil** — furrowed, deformable ground where the feet sink and slip.

- Real soil is **not uniform**: it changes from firm & grippy to soft & slippery.
- Two things vary independently:
  - **Traction** — how grippy the surface is (friction μ).
  - **Support** — how firm it is / how much the foot sinks (stiffness).
- The robot must **judge how hard the current soil is** to adapt and stay upright.

The paper's pitch

Combine traction + support into one human-readable "**soil difficulty score**" using **fuzzy logic**, and use it to drive training / reduce foot slip (>95% claimed).

What the "fuzzy soil index" actually is

A **fuzzy (Mamdani) controller**: two inputs $\rightarrow$ one difficulty score.

- Inputs: **traction** μ and **support** (stiffness).
- Each is graded *Low* / *Med* / *High* with **smooth** (fuzzy) boundaries.
- Rules $\rightarrow$ soil is *Easy* / *Moderate* / *Hard*.
- Output: one number $d \in [0.2, 0.8]$ (higher = harder).

Selling points: interpretable bins, smooth interpolation, blends two factors.

Two soil "dials"

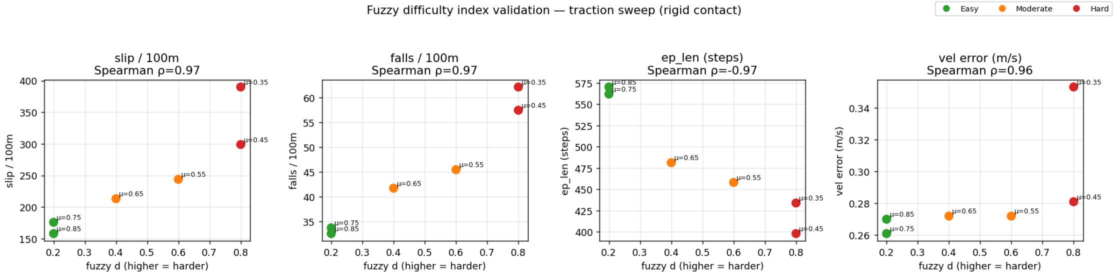
traction μ & support (firmness)

$\downarrow$ fuzzy rules $\downarrow$

difficulty = Easy / Mod / Hard

First impression: the fuzzy index looks convincing

Sweep the soil from grippy to slippery, and plot the fuzzy score d against the **real measured difficulty**:



***d* tracks difficulty almost perfectly** (slip, falls, episode length, velocity error; Spearman rank correlation $|\rho| \approx 0.97$ — same ordering of soils). **This is the chart that makes the fuzzy index look good** — but it only shows d tracks *one* axis; whether that beats the raw numbers is the next question.

Our question

Does the fuzzy index **actually add value** over just using the **raw soil numbers**?

- A fuzzy score is only useful if it **beats the raw measurements** it is built from.
- We replicate the recipe on the **Hunter** humanoid in **IsaacSim** (HumanoidVerse) and test the index four ways.
- We judge it as a **difficulty descriptor** and as a **control signal**.

Honest-negative friendly: we want the right answer, not a flattering one.

How we tested it — five settings

Study	**Simulator**	**Fuzzy used as**	**Support axis**
A	rigid PhysX	passive **label**	physically **inert**
B	rigid PhysX	**curriculum** controller	inert
C	**DFH deformable**	**descriptor**	physically **active**
① ②	DFH (+ noisy sensing)	descriptor / decisions	active
D	DFH	**redesigned** (fixed) index	active

Each study closes the previous one's loophole. "Support axis" = whether soil *firmness* actually does anything in that simulator. ①② = noisy-sensing probes on Set C's soil; **D** = we *fix* the broken rules and re-test.

Sets A & B (rigid simulator)

Set A — fuzzy as a label: does d track difficulty / beat raw μ?

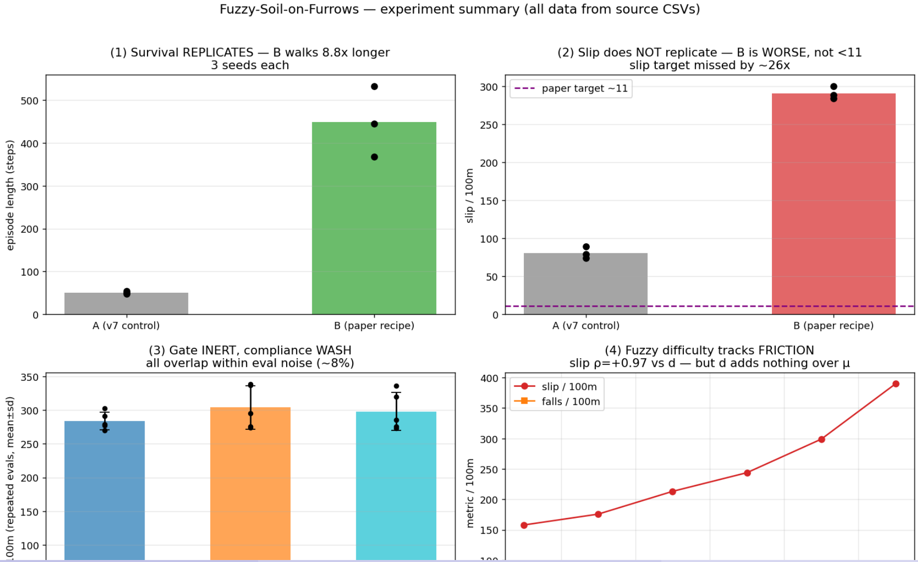
- d orders soil correctly (Spearman $\rho \approx 0.97$) . . .
- . . . but adds **nothing** over raw friction μ.
- **Why:** rigid PhysX has **no real "firmness" axis** — the foot can't sink, so the 2-dial model collapses to 1 dial (friction).

Set B — fuzzy drives the training curriculum:

- fuzzy pacing $\approx$ crisp threshold $\approx$ fixed schedule (no significant difference).

But the $\rho \approx 0.97$ chart looked so good — what's going on? $\rightarrow$

Set A/B — the full evidence (rigid simulator)



Set B: how the fuzzy *curriculum* works

Curriculum = train from **easy** soil (grippy, μ=0.80) $\rightarrow$ **hard** (slippery, μ=0.35). **The real question:** *who decides when to move to harder soil?*

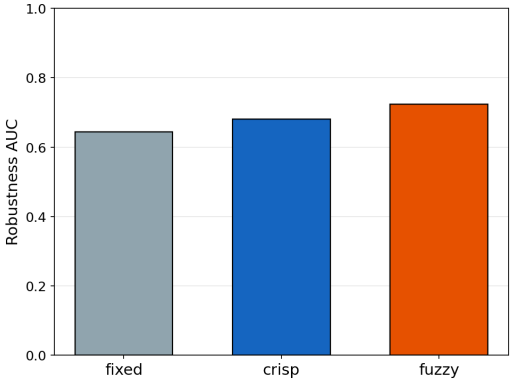
Pacing rule	Like...	Move to harder soil when...
fixed	a **timer**	a set number of blocks passed (*ignores skill*)
crisp	a **pass/fail exam**	skill $\geq$ threshold (*hard yes/no at 50%*)
fuzzy	a **progress bar**	graded scores (Hold/Maybe/Advance) **add up** to 1

Same ladder, **same** skill signal, **same** "at most 1 step harder per block" — the **only** thing that changes is the decision rule.

The honest design choice

"Skill" = **how long the robot survives** (not the soil score). Fuzzy's real twist is **accumulating a smooth score** — otherwise it would just be "crisp with a new name".

Set B: the result — fuzzy *ties* the simple rules



After training, re-test every policy across *all* soils — a fair *held-out* sweep.

Robustness AUC = one summary score = how well it survives across the board (higher = tougher everywhere):

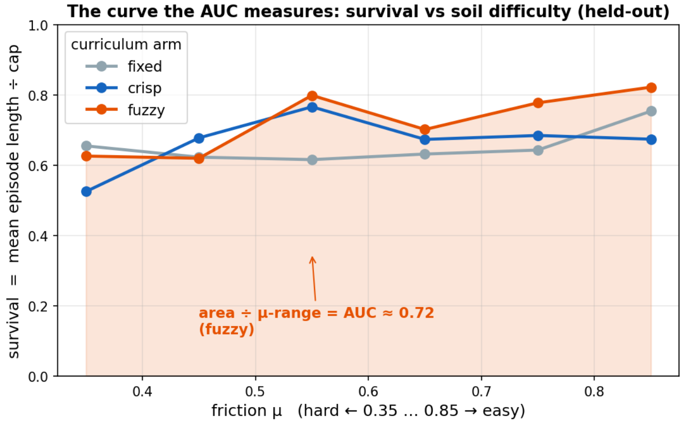
fixed **0.64** | crisp **0.68** | fuzzy **0.72**

The gap is **not significant** (Welch t=0.45, 3 seeds): fuzzy's higher average comes from **just one lucky seed**.

Bottom line

The curriculum **works**, but **fuzzy** ≈ **crisp** ≈ **fixed**: in the rigid sim the **firmness axis is dead**, so fuzzy's blending has **nothing extra to exploit**.

Set B: what the robustness AUC *measures*



After training, re-test across *all* soils $\to$ a **survival curve** (episode length $\div$ cap, vs friction μ).

Robustness AUC $=$ **shaded area** $\div$ μ-range $=$ **average survival** across soils:

$$\text{AUC} = \frac{1}{\mu_{\text{max}} - \mu_{\text{min}}} \int \frac{\overline{\text{ep\_len}(\mu)}}{\text{cap}}\, d\mu$$

(cap $=$ 1000 steps). One number per arm $=$ the bar on the previous slide.

Curves **overlap** $\Rightarrow$ AUCs tie: **.64 / .68 / .72**.

A higher curve *everywhere* $=$ tougher policy. Fuzzy's curve is **not consistently above** fixed/crisp — so it only **ties**.

Set B: the formula — how fuzzy "talks to" the policy

Key point: fuzzy is **not inside the policy**. The policy is plain PPO — fuzzy only **picks the next training soil**. One loop per training block:

1. **Train** the policy (PPO) on the current friction rung μ_{rung}.
2. **Score skill:** $m = \frac{\overline{\text{ep\_len}}}{\text{cap}} \in [0,1]$ (survives longer $\Rightarrow m \to 1$).
3. **Fuzzy** $\to$ **advance score** $a \in [0,1]$ (smooth Hold / Maybe / Advance):

$$a = \frac{0 \cdot \mu_{\text{Low}} + 0.5 \cdot \mu_{\text{Med}} + 1 \cdot \mu_{\text{High}}}{\mu_{\text{Low}} + \mu_{\text{Med}} + \mu_{\text{High}}}$$

   ($\mu_{\text{Low/Med/High}}$ = how much skill m counts as low / medium / high).
4. **Accumulate & advance:** progress $+= a$; if progress $\geq 1 \Rightarrow$ move to **harder** soil, progress $-=1$.

The 3 arms differ *only* in steps 3–4

fixed: advance on a timer (ignore m). **crisp:** advance iff $m \geq 0.5$ (hard yes/no). **fuzzy:** accumulate the graded a above.

The catch behind the pretty chart

The $\rho \approx 0.97$ looks great — but it is **only the traction (friction) axis**.

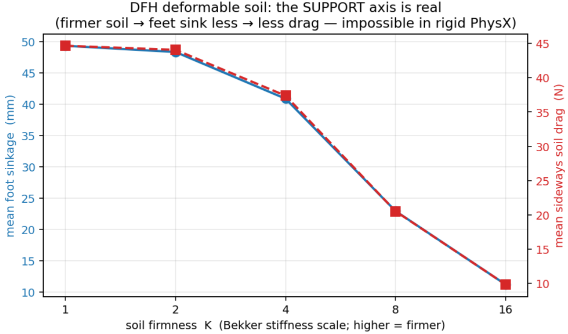
- In rigid PhysX the **support/firmness axis does nothing** (feet can't sink).
- So *d* is really just a **re-labelling of friction μ** — it carries **no information that μ doesn't already give**.
- A 2-input fuzzy model can't earn its keep when **one input is dead**.

So we need...

a simulator where soil **firmness is physically real** — next.

The fix: DFH deformable soil

DFH (Deformable Furrowed Heightfield) adds the missing physics: feet **sink** (Bekker model: compaction under the foot), generate **sideways drag**, and the soil **deforms**.



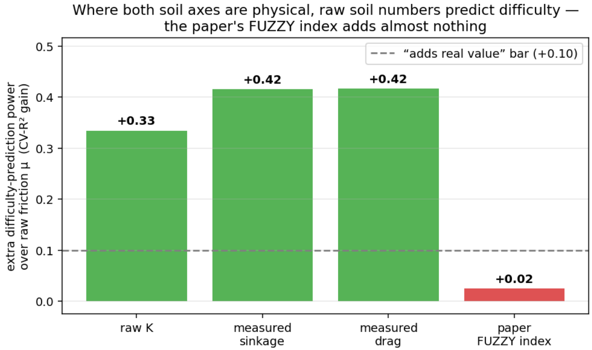
The support axis is now real:

- Soft soil ($K=1$): feet sink $\sim$**49 mm**.
- Firm soil ($K=8$): sink $\sim$**23 mm**; drag drops $\sim$**45**$\rightarrow$**20** N.
- Confirmed by a force-off control (no forces $\Rightarrow$ no effect).

Same robot, same gait — only the soil firmness changes. You can see it.

Set C: the key result

Now **both** soil dials are physical. Does the fuzzy index finally add value?



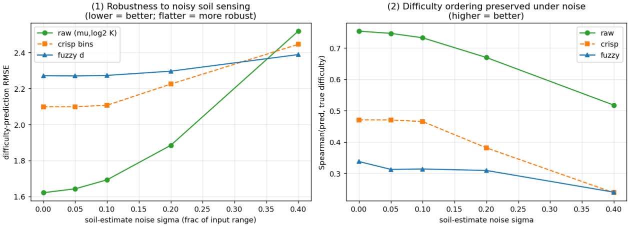
- Raw soil numbers **strongly predict** difficulty (+0.33 to +0.42).
- The paper's **fuzzy index** adds **+0.025** — essentially nothing (statistically detectable, $p = 0.001$, but practically zero).

Sharp finding

Even where firmness **is** physically active, the fuzzy mapping **fails to use it**. The problem is the **fuzzy model's structure**, **not** the simulator.

Giving fuzzy its best shot: noisy soil sensing

Fuzzy logic's real strength is **tolerating imprecise inputs**. A real robot only *estimates* the soil. So we added sensing noise.



- Fuzzy **is** noise-stable (flat line) and edges raw on error-*size only* at extreme noise.
- But it "wins" by being **stable, not accurate** — even there, raw still **orders soils better**. Flatness $\neq$ usefulness.
- **Decisions** (②): a fuzzy-driven curriculum **camps on easy soil** and never advances.

Stability of an *uninformative* signal — it doesn't swing because it barely says anything.

Set D: can we *fix* the broken fuzzy rules?

Set C blamed the **rule structure** (it ignores firmness). So we redesigned it — **only** the rule table, so support now matters at *every* traction level.

Descriptor	Uses firmness?
paper fuzzy index	**+0.025** (blind)
redesigned fuzzy	**+0.416** (sees it)
raw soil numbers	+0.33 to +0.42

Cross-validated R^2 gain over friction alone (higher = uses firmness more). The redesign lands **inside** the raw numbers' range — it *matches* them, not beats.

The fix works — as a descriptor:

- Firmness signal jumps **0.025** $\rightarrow$ **0.416** ($\sim 16\times$).
- The original failure was **structural and fixable**, not the simulator.

Honest: the fix is informed by the data's firm=easier trend — a **corrected** descriptor, not a vindication of the paper's rules. Raw soil numbers stay the ceiling; the fix only **matches** them, never exceeds.

Set D: *exactly* what we changed in the rules

Same 2 inputs, same fuzzy membership shapes, same defuzzification — we changed **only the rule lookup** (grip, firmness) → difficulty, plus the output detail (**3**→**5** levels). *(rows = grip/traction, columns = firmness)*

Paper rules (Easy / Mod / Hard)

	Soft	Med	Firm
Low	**Hard**	**Hard**	**Hard**
Mid	**Mod**	**Mod**	**Mod**
High	Mod	Easy	Easy

Low & Mid rows are flat: firmness changes **nothing** ⇒ the index is **blind** to how soft the soil is.

Redesigned rules (v2) (VE < E < M < H < VH)

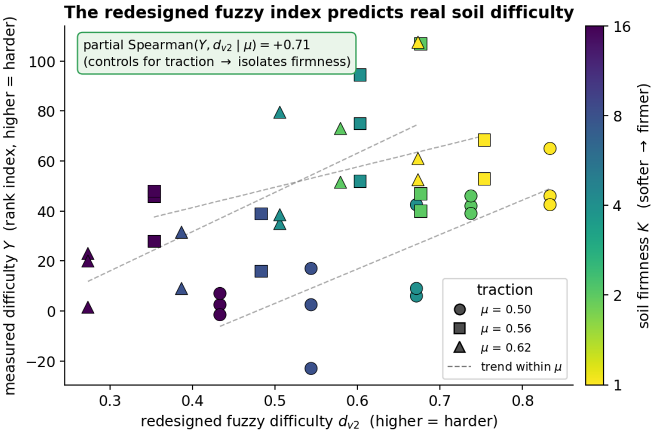
	Soft	Med	Firm
Low	**VH**	H	M
Mid	H	M	E
High	M	E	**VE**

Every row falls Soft→Firm: firmer soil = easier, at **every** grip level.

Payoff

Firmness signal jumps **+0.025** → **+0.416** ($\sim$16×) — **purely** from a better rule table. The broken part was **structural**; the fix needs **no new physics**.

Does the redesigned fuzzy index actually work? (descriptor check)



Within the measured sweep, the redesigned fuzzy index **orders real difficulty** and **separates firmness**: within each traction μ, a higher fuzzy score = harder soil (partial Spearman $+0.71$).

Descriptor validation only — d_{v2} was tuned on this regime; it aids interpretation, not a claim that fuzzy beats raw for training.

Set D: but does the fix help the robot *train*?

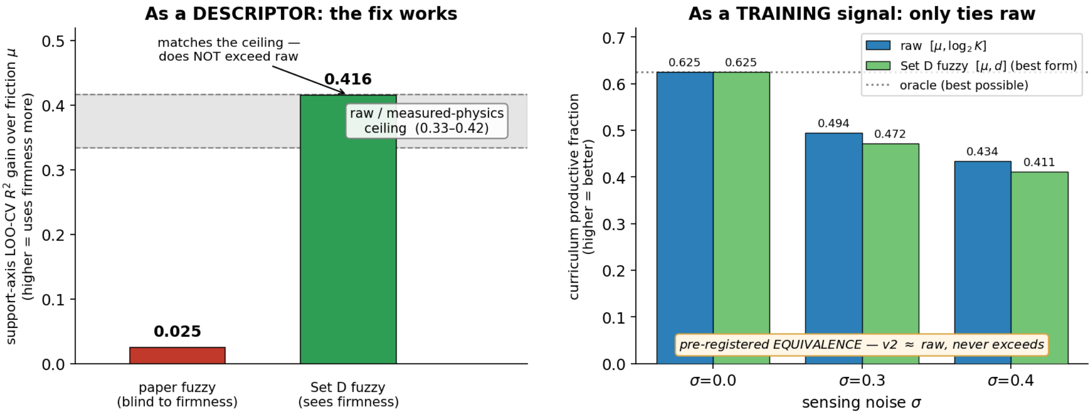
The catch: a better *description* of soil is not the same as a better *training signal*. We pace the curriculum (when to move to harder soil) with the fixed index — cheap gated proxy, vetted with an independent model.

- As a single score: **fails** — it trails the raw numbers and thrashes.
- In its **best form** $[\mu, \text{fuzzy}]$: **fully fixes** the camping and **matches** the raw numbers' accuracy...
- ...but only **ties** them — it carries the **same information** as raw $[\mu, \log_2 K]$, never beats it (a pre-registered **equivalence** result: a tie, not a win).

Decision

GPU training run not justified ($\sim$24 h saved). A better *description* of the soil does **not** become a better *training signal* than the raw numbers.

Set D in one picture: a descriptor win, not a training win



Fixing the fuzzy **rule table** makes the index physically meaningful as a **descriptor** (left: 0.025 → 0.416, reaching the raw/physics ceiling) — but as a **training** signal it only **ties** raw soil numbers (right: equivalent across noise). **Raw stays the practical ceiling.**

Verdict

Bottom line

Across rigid **and** deformable soil, the paper's fuzzy soil index shows **no demonstrable value** over the raw soil numbers — as a label, a curriculum, a descriptor, or under noisy sensing. And when we **fix** it, it becomes a good descriptor but **still only ties** raw for training.

- **Replicates:** the friction-curriculum **survival** benefit; DFH genuinely restores the firmness axis.
- **Does not add value:** the paper's mapping collapses two soil factors into a coarse score and **discards the useful information**.
- The failure is **structural** (Set C), and **fixable** (Set D) — but the fix buys **interpretability, not performance**: raw is the ceiling fuzzy can reach, not exceed.

Scope & what would change the answer

Honest scope:

- This is about **this paper's specific fuzzy mapping** — *not* all 2-D or learned terrain descriptors.
- DFH is an **uncalibrated** deformable-soil surrogate; one frozen walker.
- Not a disproof of the paper on **real** soil — a non-replication *in simulation*.

What could make "fuzzy matter":

- ~~Redesign the rules to use the firmness axis~~ — **done (Set D)**: it works as a **descriptor** but still only **ties** raw for training.
- A genuinely **new** claim with **noise-stability as the goal**, or a setting where the raw soil numbers **aren't available** (e.g. a real robot with only a coarse visual soil label, no precise sensors).
- Train a soil-robust walker (removes the single-policy caveat).

Summary

A fuzzy score is only useful if it beats the numbers it's made from.

Here it doesn't — even after we fixed it, the best it does is **tie**.

Fuzzy soil difficulty is a strong, interpretable **description** *of terrain hardness, but it does not improve curriculum* **performance** *over the raw soil parameters for Hunter locomotion.*

Code & full write-ups: `docs/experiments/fuzzy_soil_UNIFIED_conclusion.md`; models on Hugging Face `anhrisn/hunter-dfh-locomotion`.